

# Proposal: A Test for Pre-Causal Resonance (PCR) in LHC Data

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## Abstract

Event-State Theory (EST) proposes that physical events are not instantaneous points in time, but phase transitions (crystallizations) preceded by a convergence of potential information. This model predicts a specific, falsifiable signature: **Pre-Causal Resonance (PCR)**. We propose a re-analysis of existing LHC "Minimum Bias" data to detect correlated vacuum fluctuations in the nanoseconds prior to high-energy collision triggers.

## The Hypothesis

Standard Quantum Field Theory treats the vacuum state at  $t < 0$  (pre-collision) as uncorrelated noise. EST posits that the "Field of Potential" must increase in information density to satisfy the Cost Function for the impending event.

- **Prediction:** The "Background Noise" in the detector calorimeters will show a statistically significant, non-random "inward ripple" pattern centering on the vertex coordinate  $x, y, z$  at  $t < 0$ .

## Proposed Methodology

We request a study of **Zero Bias** or **Minimum Bias** datasets (e.g., CMS Run 2 or 3).

1. **Target:** Identify high-multiplicity collision events.
2. **Window:** Analyze the raw calorimeter voltage spikes in the window  $t = [-10ns, 0ns]$ .
3. **Filter:** Search for concentric or convergent signal gradients that correlate with the subsequent event magnitude.

## Falsifiability Condition

If the pre-event vacuum state remains statistically indistinguishable from random Gaussian noise at a sensitivity cross-section of  $\sigma_{PCR} > [\text{Threshold}]$ , the dynamical mechanism of EST is falsified.